

KARTHIK MANNEM

Los Angeles, CA

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SUMMARY

Machine Learning Engineer who approaches problem-solving as an end-to-end AI pipeline, ingesting complex real-world inputs, performing structured feature engineering, training and fine-tuning deep learning and transformer-based architectures, and deploying scalable inference systems. With 3+ years of experience across predictive modeling, computer vision, and LLM-powered Retrieval-Augmented Generation (RAG) workflows, I focus on evaluation rigor, performance optimization, and production-grade deployment.

PERSONAL PROJECTS

•Preventive Care Recommendation Agent | LLM + RAG (LangChain)

LLM-powered healthcare decision support system deployed publicly.

- Designed and deployed an LLM-powered preventive care recommendation system that analyzes patient age, medical history, and risk factors to generate personalized screening and vaccination guidance aligned with CDC guidelines.
- Architected a Retrieval-Augmented Generation (RAG) pipeline using LangChain, implementing document chunking, embeddings, and vector-based semantic retrieval.
- Applied prompt engineering techniques to improve reasoning accuracy and reduce hallucinated medical outputs.
- **Tech:** Python, LangChain, OpenAI API, Vector Search, Flask, Docker

•SplitWise Pro – AI-Powered Expense Sharing Platform

Full-stack expense management system with OCR-based receipt parsing and cloud deployment.

- Built a full-stack expense sharing platform for group expense tracking, receipt uploads, balance calculation, and optimized settlement generation.
- Integrated Firebase Authentication with a Node.js/Express backend to secure protected routes and synchronize authenticated users with MongoDB Atlas.
- Implemented AWS S3-based receipt storage with Google Vision API and OpenAI-assisted OCR parsing to extract structured expense details.
- **Tech:** Next.js, TypeScript, Node.js, Express, MongoDB Atlas, Firebase Auth, AWS S3, Google Vision API, OpenAI API

•Estimating Product Length using BERT on Amazon Dataset

NLP-based regression model for structured e-commerce prediction.

- Fine-tuned a pre-trained BERT model using product titles, descriptions, and category metadata to predict product length in a large-scale e-commerce dataset.
- Implemented text preprocessing, tokenization, and feature engineering pipelines to improve model convergence and prediction stability.
- Applied log transformation and model ensembling techniques to enhance regression performance and reduce prediction variance.
- **Tech:** Python, BERT, HuggingFace Transformers, Scikit-learn, Pandas

•Fake Currency Detection Using CNN

Computer vision-based counterfeit currency detection system.

- Developed a CNN-based image classification model using TensorFlow and OpenCV to detect counterfeit currency from a dataset of 10,000+ images.
- Implemented image preprocessing and data augmentation techniques to improve robustness and reduce false positives.
- Optimized training and evaluation workflows to enhance generalization across varied lighting and image conditions.
- **Tech:** Python, TensorFlow, OpenCV, CNN, NumPy

TECHNICAL SKILLS

Programming: Python, Java

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV

NLP & LLM Systems: Transformers, HuggingFace, LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG)

Data Processing: Pandas, NumPy, Feature Engineering, Data Preprocessing

Deployment & Tools: Docker, Flask, REST APIs, Git, GitHub

Databases: MongoDB, PostgreSQL, MySQL

Cloud: AWS (EC2, S3)

EDUCATION

- California State University, Northridge

Master of Science in Computer Science | CGPA: 3.9 / 4.0

Los Angeles, CA
Aug 2024 – May 2026 (Expected)
- Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.21 / 10.00

Chennai, India
Aug 2020 – May 2024

EXPERIENCE

- AI Model Evaluation Analyst

Handshake AI

Sep 2025 – Present

 - Evaluated 1,000+ large language model (LLM) outputs using structured quality benchmarks to assess response accuracy, coherence, and factual consistency.
 - Identified recurring model failure cases and categorized hallucination patterns, contributing to improved prompt refinement workflows and reducing evaluation inconsistencies by 20%.
 - Collaborated with cross-functional teams to provide systematic feedback for improving model behavior across multiple domain-specific evaluation tasks.
 - Maintained high annotation agreement standards across large datasets, increasing benchmarking reliability and validation consistency.
- Machine Learning Engineer (Freelance)

Self-Employed

Aug 2023 – Present

 - Designed and deployed deep learning models (LSTM, CNN) for 3+ client applications, delivering production-ready inference pipelines.
 - Built end-to-end ML workflows including data preprocessing, feature engineering, model training, and cross-validation, improving model training efficiency by 30%.
 - Containerized ML services using Docker and exposed REST APIs via Flask, enabling scalable real-time prediction systems with reduced deployment overhead.
 - Optimized data pipelines and training configurations, reducing overall training time by 25% and improving inference latency in deployed systems.

POSITIONS OF RESPONSIBILITY

- Board of Directors – Facilities and Operations Specialist

University Student Union, CSUN

Aug 2024 – Present

 - Oversaw campus infrastructure initiatives impacting 38,000+ students and managed a \$5M operational budget.
 - Collaborated with executive leadership to improve sustainability, accessibility, and space utilization across university facilities.
- Content Lead

Google Developer Student Clubs

Aug 2022 – Apr 2024

 - Led a team of 8+ members, organizing workshops and publishing technical content for 200+ participants.
 - Mentored contributors in Git/GitHub workflows and coordinated cross-campus technical collaborations.
- Student Volunteer

National Service Scheme (NSS)

2021 – 2024

 - Organized blood donation drives, campus clean-ups, and environmental awareness campaigns impacting 1000+ community members.
 - Led donation drives supporting orphanages and local social programs.

LICENSES AND CERTIFICATIONS

Artificial Intelligence and Machine Learning (Foundation and Associate) – AWS
Fundamentals of Deep Learning – NVIDIA
Program Essentials in Python Programming (Intermediate Level) – CISCO
Model Validation 2 Expert – Handshake AI